

Tanhim Malik

tanhimmalik321@gmail.com | New York, NY

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PERSONAL SUMMARY

Software engineer with experience building internal developer platforms, cloud automation, trading applications, and observability systems using TypeScript, React, Java, Python, and AWS. Focused on reducing operational friction — from cutting incident detection time by 60% to automating infrastructure provisioning — across fintech and enterprise environments.

WORK EXPERIENCE

Apprentice Software Engineer

Tripadvisor, New York, NY

March 2024 - October 2024

- Built internal developer platform features as **TypeScript** and **React** plugins for **Cortex**, automating service creation, resource management, and engineering workflows.
- Developed a **GitLab-to-GitHub Migration Assistant** with API integrations, asynchronous status polling, validation, and error handling for self-service repository migrations.
- Automated **AWS infrastructure provisioning** and deployment workflows with **AWS CDK** and **TypeScript** across monitoring, scaffolding, and migration tool stacks.
- Converted service-catalog workflows from file-based processes to bulk **Cortex API submissions**, helping achieve **100% documented ownership** across cataloged products.
- Used LLM-assisted workflows to accelerate refactoring of Cortex plugin code and generate onboarding documentation, cutting new-plugin setup time for engineering teams.

Tech Stack: **TypeScript, React, AWS (CDK, S3, EC2), Node.js, Cortex, Git, Docker**

Software Engineer Intern

Jefferies Group, New York, NY

February 2023 - August 2023

- Developed and supported **Java** applications for the ION Trading Platform, integrating custom APIs to improve trader workflows and system functionality.
- Developed a risk aggregation application for mortgage trading desks using **Java, Spring Boot**, and **Kafka**, supporting the COO, CFO, and dozens of traders.
- Built an internal observability platform using **Python, Prometheus, Grafana, Loki**, and **Promtail**; delivered seven dashboards and critical alerts that reduced incident detection time by **60%**.
- Automated kdb+ data extraction and analysis workflows with **Python** and **Unix scripting**, reducing analysis time by more than **85%**.
- Contributed to the migration of **kdb+ databases** from on-premises infrastructure to **AWS EC2** and **S3**, improving query performance by up to **40%**.

Tech Stack: **Java, Spring Boot, Kafka, Python, Grafana, Prometheus, Loki, Promtail, AWS (EC2, S3), kdb+, Unix**

SKILLS

- **Languages & Frameworks:** JavaScript, Java, Python, HTML/CSS, SQL, TypeScript, React, Node.js, Spring Boot, PostgreSQL, Flask, kdb+, Kafka, R
- **Cloud & Tools:** Git, AWS (EC2, S3, Lambda, Redshift, CDK), Docker, GitHub, Prometheus, Grafana

- **AI:** Generative AI, LLMs, Prompt Engineering, AI-Driven Development

PROJECTS

- **E-Commerce Admin Dashboard** | [Demo](#)
 - Built a component-driven React dashboard with dynamic data tables supporting real-time CRUD operations across catalog, inventory, and order management views.
 - Implemented persistent authentication and theming with React Hooks and custom state management, reducing reliance on external state libraries.

EDUCATION

Western Governors University

Bachelor of Science in Computer Science

In Progress

Year Up New York

Technical Diploma in Software Development

August 2023